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12. What is the difference between a process that is ready and a process that is waiting?

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12. A process that is ready could make progress if given a time slice, but giving a time slice to a process that is waiting would merely waste time since it cannot progress until some event occurs.

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13. What is the difference between virtual memory and main memory?

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13. Virtual memory is the memory space whose presence is merely simulated by swapping blocks of data back and forth between a disk and the memory actually present in the machine.

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18. What is a context switch?

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18. The machine begins by executing a program, called the bootstrap, at a predetermined location in memory. This program directs the machine to load a program (the operating system) from mass storage into main memory. The original program tells the machine to transfer its attention to the program just loaded.

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25. A process is said to be I/O-bound if it requires a lot of I/O operations, whereas a process that consists of mostly computations within the CPU/memory system is said to be compute-bound. If both a compute-bound process and an I/O-bound process are waiting for a time slice, which should be given priority? Why?

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25. A process is said to be I/O-bound if it requires a lot of I/O operations, whereas a process that consists of mostly computations within the CPU/memory system is said to be compute-bound. If both a compute-bound process and an I/O-bound process are waiting for a time slice, which should be given priority? Why?

25. The I/O-bound process. This allows the controllers to start with the I/O activities. Then the compute-bound process can run while the other is waiting for these slower activities to take place. As a general rule of thumb, priority should be given to the slower activity.

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27. Write a set of directions that tells an operating system's dispatcher what to do when a process's time slice is over.

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27. Save the current process' state;

select another process from the process table;

load that process' state;

start the next time slice.

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30. What is meant by an interrupt handler in multiprogramming systems and what is its significance?

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30. What is meant by an interrupt handler in multiprogramming systems and what is its significance?

30. First: Interrupt signal occurs.

Second: Machine completes its current instruction.

Third: Machine saves the current program state.

Fourth: Machine begins executing the interrupt routine.

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***39.** Why is disabling the interrupts in a multicore operating system not considered to be an efficient approach?

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*39. Why is disabling the interrupts in a multicore operating system not considered to be an efficient approach?

39. As the processes producing the printed material terminate, their output that has accumulated in mass storage is placed in a queue to wait for the printer. Each time the printer finishes the output of a process, it begins printing the next unit of output in this queue.

51. What are the conditions that lead to a deadlock?

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51. What are the conditions that lead to a deadlock?

Conditions required for deadlock

1. Competition for non-sharable resources
2. Resources requested on a partial basis
3. An allocated resource can not be forcibly retrieved